Validation and Failure Testing

Nine failure tests run during continuous pings, with procedures, results and screenshots.

Blake Otto · Catalyst Switch Lab — Small Office Network Build
github.com/blakeotto57/Catalyst-Switch-Lab-Small-Office-Network-Build

Failure tests

Each test ran during a continuous ping from the test laptop on SW2. Because the laptop and the server VM shared one network adapter, most tests pinged SW3's management address (10.10.99.3), which follows the same path through R1.

#	Test	Result
1	One link in SW2's EtherChannel fails	Pass — Po1 stayed up on one link; no pings lost
2	SW2 loses both core uplinks	Pass — SW3 Fa0/4 went blocking → listening → learning → forwarding; ~40 s outage (10 timeouts × Windows' 4 s default timeout) because SW3 runs 802.1D
3	SW2's uplinks are restored	Pass — Po1 re-formed and became SW2's root port again; SW3 Fa0/4 returned to blocking
4	Cross-link role check	Pass — SW3 Fa0/4 blocking, SW2 Gi0/4 designated forwarding
5	BPDU received on an edge port	Pass — %SPANTREE-2-BLOCK_BPDUGUARD and %PM-4-ERR_DISABLE; test port err-disabled
6	Native VLAN mismatch	Pass — %CDP-4-NATIVE_VLAN_MISMATCH logged; a later accidental mismatch (VLAN 1 vs 900 after SW3 reloaded) was also caught by spanning tree PVID-inconsistency blocking on both switches
7	Root bridge change	Pass — SW2 became root for VLAN 10, the blocked port moved from SW3 Fa0/4 to SW3's uplink (cross-link cost 19 < 12 + 12 via R1), and both returned when the priority was restored
8	Lab device attempts to reach a home device	Pass for the block — ACL line 10 hit counters and %SEC-6-IPACCESSLOGDP; lab-internal traffic unaffected. The "internet still works" part could not be tested (see requirement 10 limitation)
9	Second device on a secured port (optional)	Pass — %PORT_SECURITY-2-PSECURE_VIOLATION naming the second MAC; port err-disabled and recovered with shutdown / no shutdown

Test 1: One link in SW2's EtherChannel fails

What it proves: EtherChannel redundancy
Pass criteria: Traffic continues over the remaining link and the bundle stays up
Result: Pass — Po1 stayed up on one link; no pings lost

Procedure and evidence

Initially on SW2 we see this etherchannel summary:

```
SW2#sh ether sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

SW2#
```

Initially on R1 we see this etherchannel summary:

```
R1#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

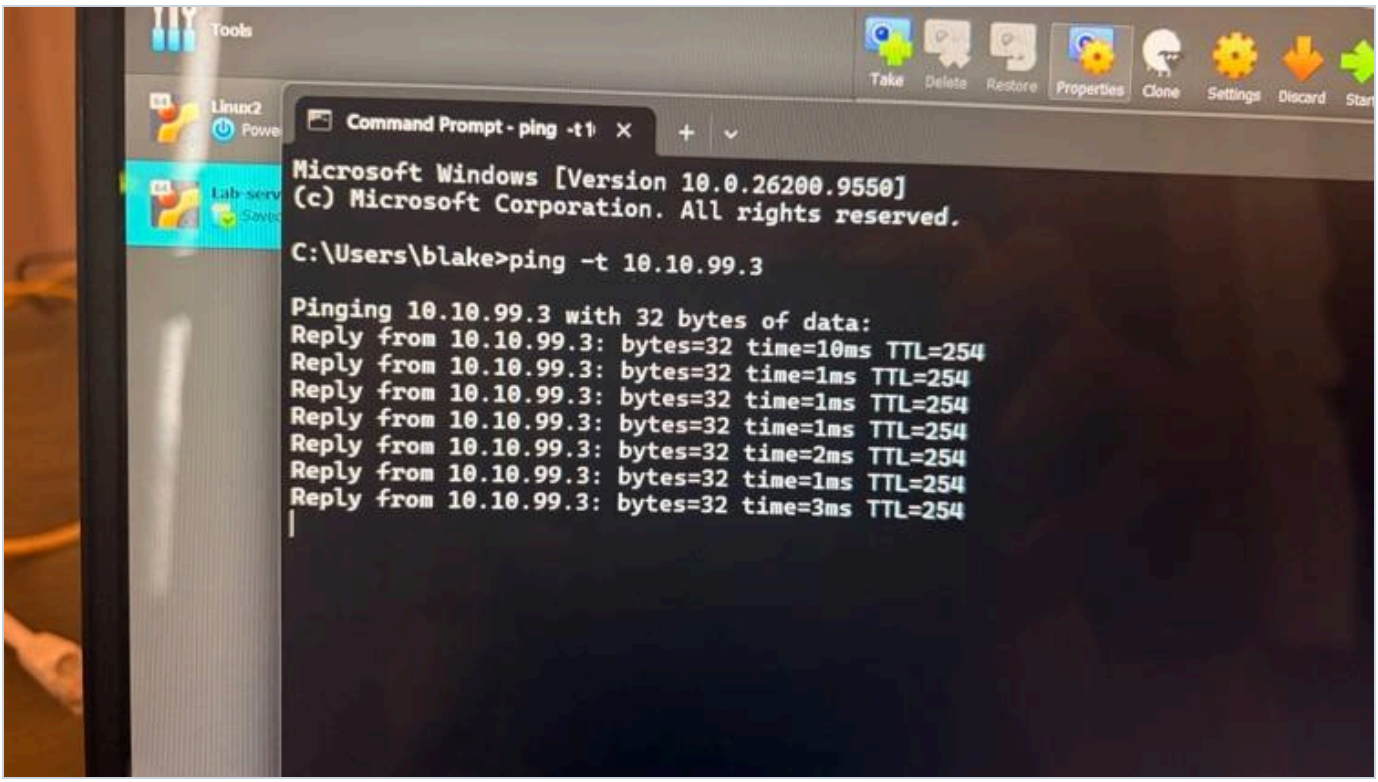
        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

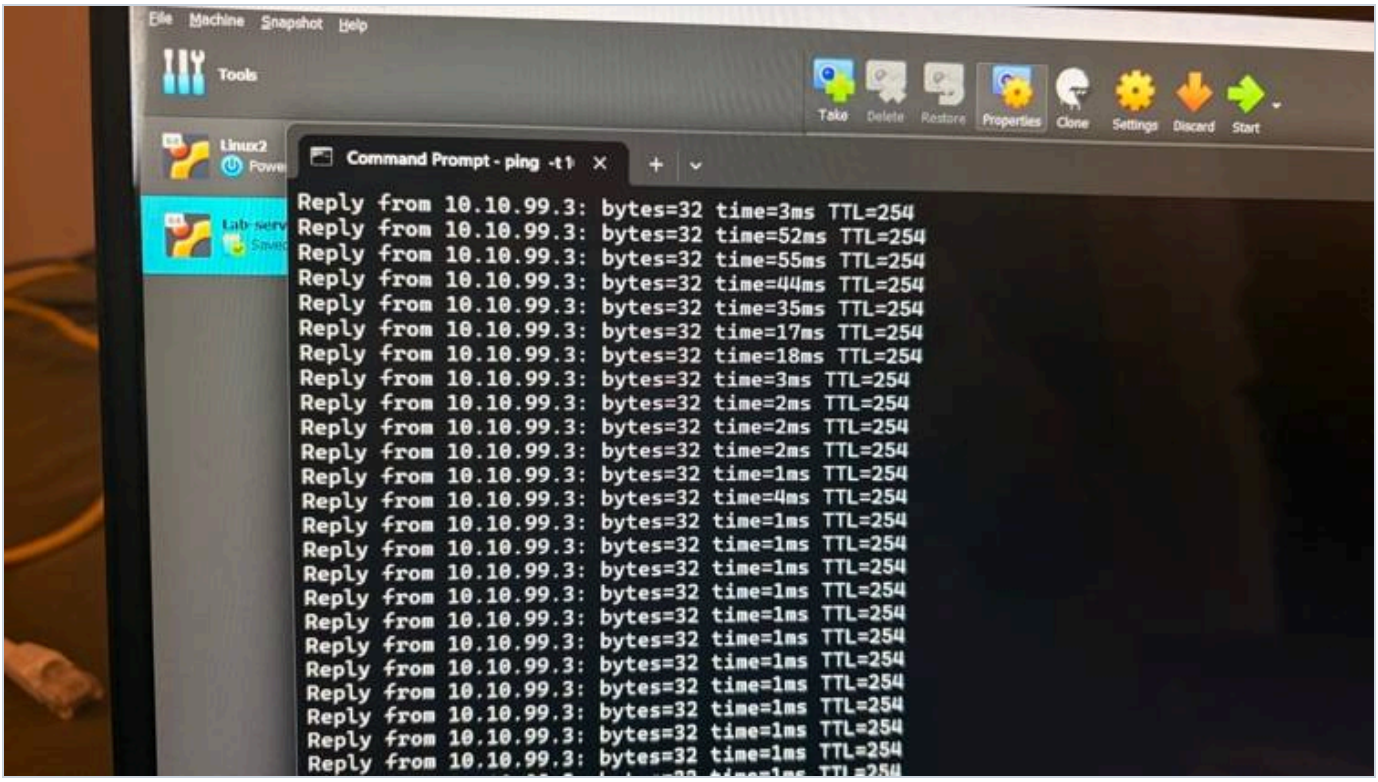
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------
1      Po1(SU)         LACP      Fa0/1(P)    Fa0/2(P)
2      Po2(SU)          -        Fa0/3(P)    Fa0/4(P)

R1#
```

Now on the test laptop connected to SW2 g0/3 I start pinging SW3's management IP "ping -t 10.10.99.3"



I then unplugged G0/1 on SW2 and there were still replies.



Then I checked the etherchannel summary on SW2:

```
SW2#sh eth su
*Mar  1 03:40:30.948: %SYS-5-CONFIG_I: Configured from console by blake on console
SW2#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Gi0/1(D)    Gi0/2(P)

SW2#
```

As we can see the etherchannel is still formed even if one link is down. I am still getting ping replies.

I put the G0/1 cable back in and ran eth sum and it is restored so traffic still continued even if a link failed.

```
SW2#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

SW2#
```

Test 2: SW2 loses both core uplinks

What it proves: Backup path through the cross-link
Pass criteria: Spanning tree unblocks the cross-link and traffic reaches the core through SW3; convergence time is recorded
Result: Pass — SW3 Fa0/4 went blocking → listening → learning → forwarding; ~40 s outage (10 timeouts × Windows' 4 s default timeout) because SW3 runs 802.1D

Procedure and evidence

On SW2:

```
SW2#sh span

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p
```

```
SW2#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

On SW3:

```
SW3#sh span

Spanning tree 1 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6200
  Configured hello time 2, max age 20, forward delay 15
  We are the root of the spanning tree
  Topology change flag not set, detected flag not set, changes 2
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 1, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 1 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 32768, address 0007.0ecc.6200
   Designated bridge has priority 32768, address 0007.0ecc.6200
   Designated port is 2, path cost 0
   Timers: message age 0, forward delay 0, hold 0
   BPDU: sent 6738, received 0

Interface Fa0/4 (port 16) in Spanning tree 1 is FORWARDING
   Port path cost 19, Port priority 128
   Designated root has priority 32768, address 0007.0ecc.6200
   Designated bridge has priority 32768, address 0007.0ecc.6200
   Designated port is 16, path cost 0
   Timers: message age 0, forward delay 0, hold 0
   BPDU: sent 3600, received 0

 --More--
```

```
SW3#show spanning-tree vlan 10

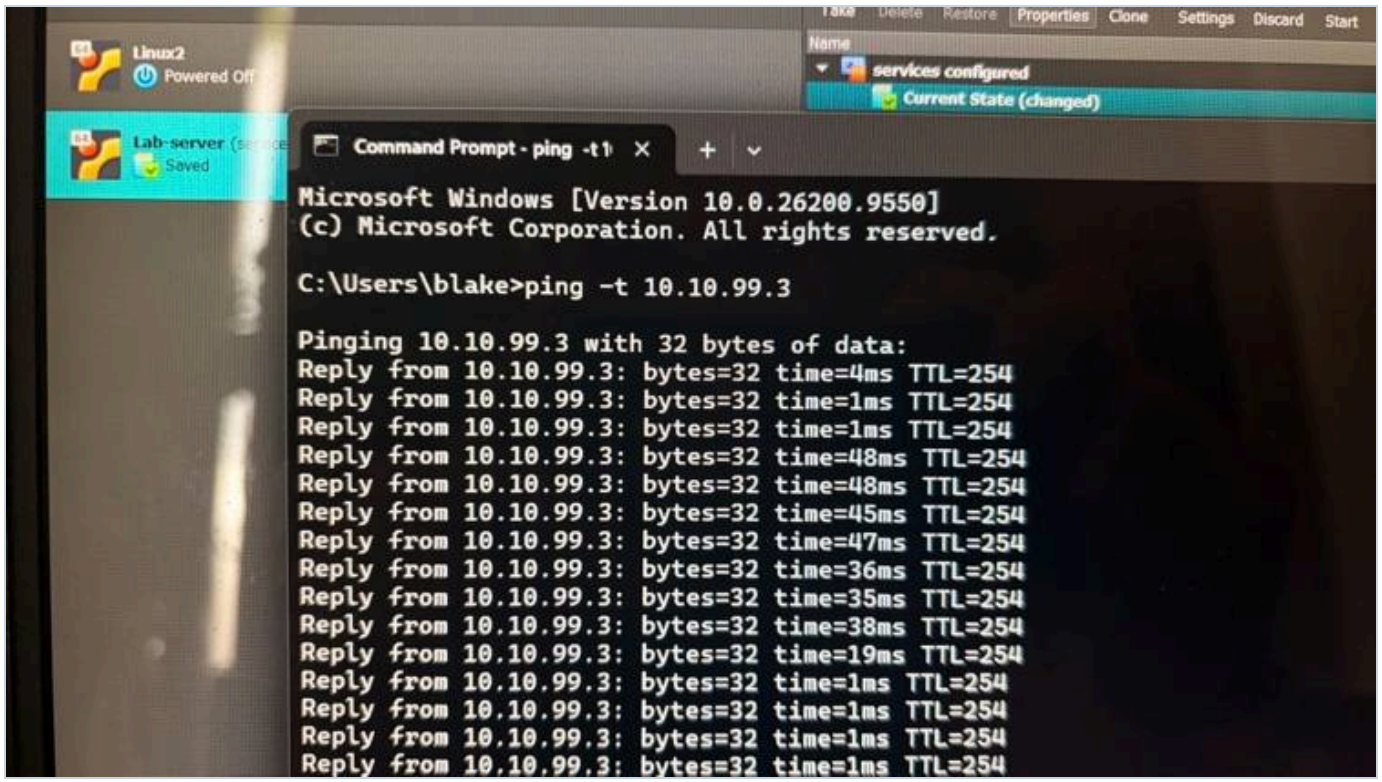
Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 7
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 5, received 6800

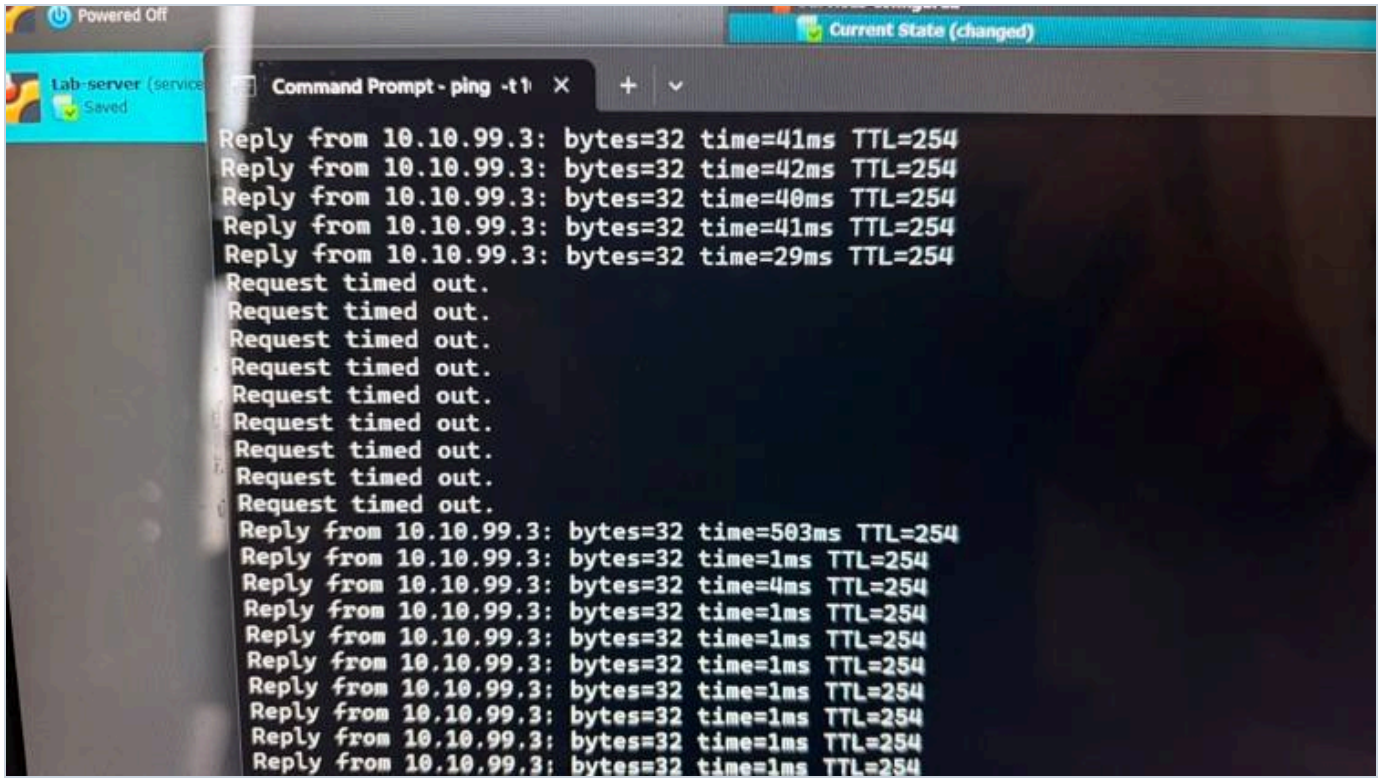
Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 4, forward delay 0, hold 0
   BPDU: sent 89, received 3572
SW3#
```

We can see that SW3's f0/4 is blocked in VLANS due to STP.

To test I connected my laptop from SW2 to ping 10.10.99.3 which were successful then I unplugged the etherchannel (G0/1-2) on SW2 to see if it would use the backup link on SW2's G0/4.
Before SW2 G0/1-2 unplug:



After output when EtherChannel was unplugged:



You can see it was blocked then SW3's f0/4 was listening and learning then started forwarding. Now you can see f0/4 is forwarding now that po1 is down.

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 8
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 8, received 7292

Interface Fa0/4 (port 16) in Spanning tree 10 is FORWARDING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 32768, address 0007.0ecc.6201
   Designated port is 16, path cost 12
   Timers: message age 0, forward delay 0, hold 0
   BPDU: sent 150, received 4006
SW3#
```

Then I replugged the ports back into po1 and we can see SW2's etherchannel is back up.

```
SW2#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

SW2#
```

and SW3's f0/4 is back to blocking:

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 9
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 9, received 7387

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
 --More--
```

on SW2, po1 is back up and is the root.

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

Test 3: SW2's uplinks are restored

What it proves: Recovery to the normal path
Pass criteria: Traffic returns to SW2's own uplinks and the cross-link blocks again
Result: Pass — Po1 re-formed and became SW2's root port again; SW3 Fa0/4 returned to blocking

Procedure and evidence

After restoring SW2's uplinks (end of test 2): SW2's EtherChannel re-formed with both links bundled:

```
SW2#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

SW2#
```

SW2 uses Po1 as its root port again and the cross-link Gi0/4 is designated forwarding:

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

SW3’s Fa0/4 returned to blocking:

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 9
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 9, received 7387

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
 --More--
```

Test 4: Cross-link role check

What it proves: Spanning tree design
Pass criteria: The cross-link port on SW3 is blocking while SW2's side is forwarding
Result: Pass — SW3 Fa0/4 blocking, SW2 Gi0/4 designated forwarding

Procedure and evidence

This output on SW2 shows that SW2's cross link is forwarding while SW3's side is blocking:

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 9
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 9, received 7387

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
 --More--
```

Test 5: BPDU received on an edge port

What it proves: Loop protection
Pass criteria: The port is error-disabled and the event is logged
Result: Pass — %SPANTREE-2-BLOCK_BPDUGUARD and %PM-4-ERR_DISABLE; test port err-disabled

Procedure and evidence

Before tests we can see port security enabled on SW2:



I will configure an unused edge port on SW2 to test this:
int g0/5
desc BPDU-GUARD-TEST
sw mode acc
sw acc vlan 10
spanning-tree portfast
spanning-tree bpduguard en
no shut

I then connected my test laptop to SW2's g0/3, connected a cable from SW2's g0/5 to SW3's f0/5 and started ping to SW3's management vlan, "ping -t -w 1000 10.10.99.3"
The pings were successful.
Then on SW 2 I ran:
show interfaces status err-disabled
show logging | i BPDUGUARD|ERR_DISABLE

and got this output:

```
SW2#show interfaces status err-disabled

SW2#show logging | i BPDUGUARD|ERR_DISABLE
*Mar  1 02:48:33.649: %PM-4-ERR_DISABLE: psecure-violation error detected on Gi0/3, putting Gi0/3 i
n err-disable state
SW2#
```

Since nothing came up for g0/5 I ran these on SW2:

```
SW2#show interfaces status | i Gi0/5
Gi0/5     BPDU-GUARD-TEST    notconnect   10           auto   auto 10/100/1000BaseTX
Gi0/50                       disabled     999          auto   auto Not Present
Gi0/51                       disabled     999          auto   auto Not Present
Gi0/52                       disabled     999          auto   auto Not Present
SW2#show run interface g0/5
Building configuration...

Current configuration : 172 bytes
!
interface GigabitEthernet0/5
 description BPDU-GUARD-TEST
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
end
```

While doing this I realized f0/5 on SW3 is still administratively disabled so I did “no shut” on it and ran the logging commands again on SW2 and got the correct output:

```
SW2#show interfaces status err-disabled

Port      Name               Status       Reason               Err-disabled Vlans
Gi0/5     BPDU-GUARD-TEST    err-disabled bpduguard
SW2#show logging | i BPDUGUARD|ERR_DISABLE
*Mar  1 04:27:42.707: %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port Gi0/5 with BPDU Guard enab
led. Disabling port.
*Mar  1 04:27:42.707: %PM-4-ERR_DISABLE: bpduguard error detected on Gi0/5, putting Gi0/5 in err-di
sable state
SW2#
```

so we see that when a BPDU got sent to an edge port(g0/5), SW2 shut the port down

Test 6: Native VLAN mismatch

What it proves: Trunk misconfiguration detection
Pass criteria: The mismatch is detected and logged by the switches
Result: Pass — %CDP-4-NATIVE_VLAN_MISMATCH logged; a later accidental mismatch (VLAN 1 vs 900 after SW3 reloaded) was also caught by spanning tree PVID-inconsistency blocking on both switches

Procedure and evidence

Initial viewing of VLANs and cdp status on SW2:

```
SW2#sh cdp neigh g0/4
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
SW3.blakeotto.com
                 Gig 0/4           162           T S      WS-C3548- Fas 0/4
SW2#h int g0/4 sw | i Native
help int g0/4 sw | i Native
     ^
% Invalid input detected at '^' marker.

SW2#show int g0/4 sw | i Native
Trunking Native Mode VLAN: 900 (NATIVE)
Administrative Native VLAN tagging: enabled
Administrative private-vlan trunk Native VLAN tagging: enabled
SW2#
```

We see CDP is enabled so if there is vlan mismatch we will get alerts.

Now to test on my laptop I pinged “ping -t -w 1000 10.10.99.3” to start pinging the VLAN.

Then I created the mismatch on SW via:
int g0/4
sw trunk native vlan 99
end

Then we wait for CDP to detect the mismatch and run native vlan mismatch command:
show logging | i NATIVE|PVID|INCONSISTENT

and see CDP detected it:

```
SW2#show logging | i NATIVE|PVID|INCONSISTENT
*Mar  1 04:34:19.161: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthern
et0/4 (99), with SW3.blakeotto.com FastEthernet0/4 (900).
SW2#show logging | i NATIVE|PVID|INCONSISTENT
```

Then I restored the native vlan trunk back to 900

```
SW2(config)#int g0/4
SW2(config-if)#sw trunk nat
SW2(config-if)#sw trunk native vla
SW2(config-if)#sw trunk native vlan 900
SW2(config-if)#end
SW2#
*Mar  1 04:36:14.739: %SYS-5-CONFIG_I: Configured from console by blake on console
SW2#show int g0/4 sw | i Native
Trunking Native Mode VLAN: 900 (NATIVE)
Administrative Native VLAN tagging: enabled
Administrative private-vlan trunk Native VLAN tagging: enabled
SW2#
```

Test 7: Root bridge change

What it proves: Spanning tree control
Pass criteria: The root moves as configured, the blocked port changes accordingly, and both return when priorities are restored
Result: Pass — SW2 became root for VLAN 10, the blocked port moved from SW3 Fa0/4 to SW3's uplink (cross-link cost 19 < 12 + 12 via R1), and both returned when the priority was restored

Procedure and evidence

We see the initial spanning tree of vlan 10 on SW2 and SW3:

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 2
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 1, received 413

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 3, forward delay 0, hold 0
   BPDU: sent 1, received 34
SW3#
```

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

Now to test, on SW2 I will move the root by changing its priority “spanning-tree vlan 10 pri 20480” and now we see SW2 is the root for VLAN 10:

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    20490
             Address     0019.e885.9000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    20490  (priority 20480 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Desg FWD 12        128.56   P2p


SW2#
```

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 20490, address 0019.e885.9000
  Root port is 16, cost of root path is 19
  Topology change flag not set, detected flag not set, changes 3
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is BLOCKING
   Port path cost 12, Port priority 128
   Designated root has priority 20490, address 0019.e885.9000
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 12
   Timers: message age 3, forward delay 0, hold 0
   BPDU: sent 2, received 525

Interface Fa0/4 (port 16) in Spanning tree 10 is FORWARDING
   Port path cost 19, Port priority 128
   Designated root has priority 20490, address 0019.e885.9000
   Designated bridge has priority 20490, address 0019.e885.9000
   Designated port is 4, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 2, received 146
SW3#
```

```
R1#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    20490
             Address     0019.e885.9000
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     0016.9d94.0980
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po1                 Root FWD 12        128.56   P2p
Po2                 Desg FWD 12        128.64   P2p Peer(STP)


R1#
```

and we see R1 is no longer the root but is the root for po1, SW3 f0/4 was blocking and is not forwarding, and SW3's f0/2 is now blocking.

Now to restore I will do this on SW2:
"spanning-tree vlan 10 priority 28672" and now we can see that SW2 is no longer the root bridge.

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

and SW3 and R1 have been restored back to their original STP roles:

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 4
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 1, forward delay 0, hold 0
   BPDU: sent 5, received 648

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 4, forward delay 0, hold 0
   BPDU: sent 3, received 267
SW3#
```

```
R1#sh span
R1#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     0016.9d94.0980
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po1                 Desg FWD 12        128.56   P2p
Po2                 Desg FWD 12        128.64   P2p Peer(STP)

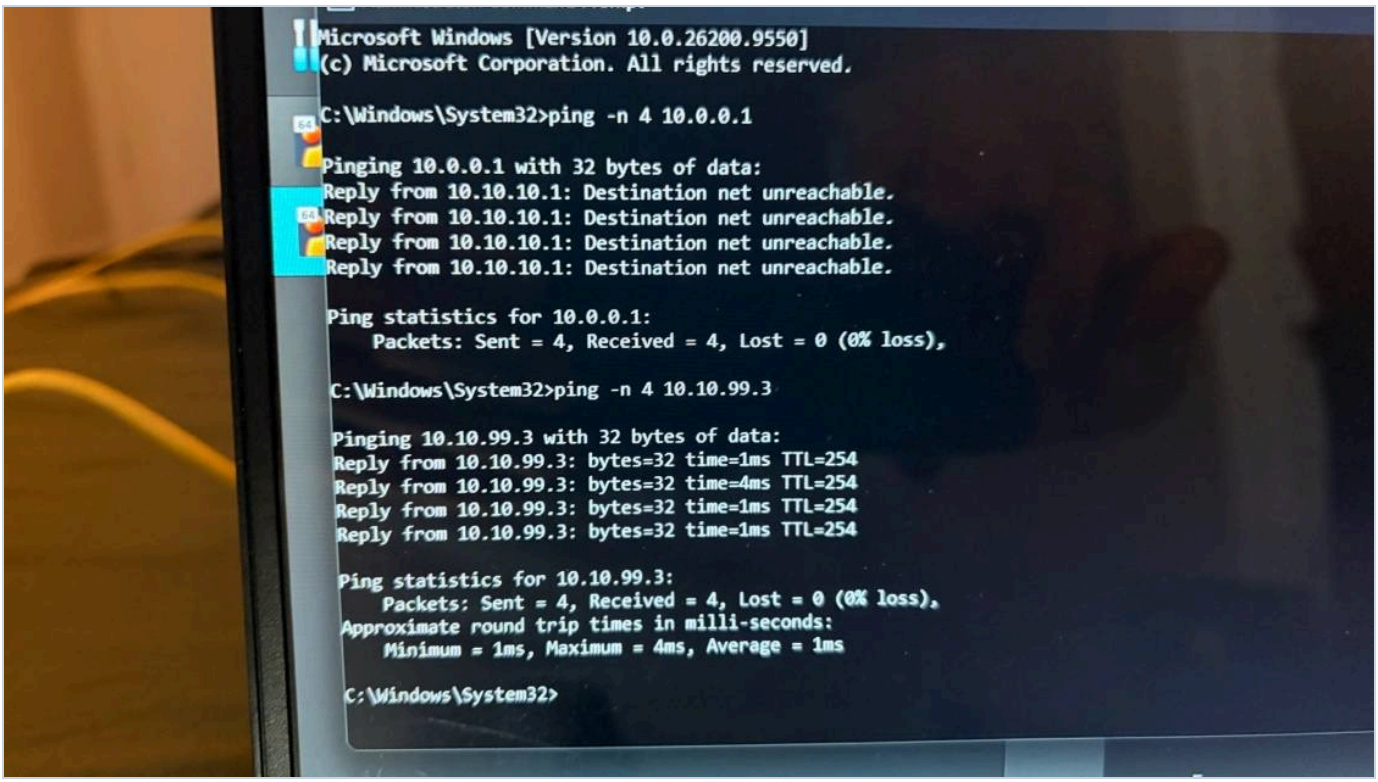

R1#
```

Test 8: Lab device attempts to reach a home device

What it proves: Access list enforcement
Pass criteria: The connection is blocked while internet access still works
Result: Pass for the block — ACL line 10 hit counters and %SEC-6-IPACCESSLOGDP; lab-internal traffic unaffected. The “internet still works” part could not be tested (see requirement 10 limitation)

Procedure and evidence

On R1 I will clear all access list counters before the test via “clear access-list counters LAB_BLOCK_HOME”
Then on the test laptop on SW2 I will ping the home router which should fail then ping the management vlan which should succeed.



Then on R1 I checked the ip access logs and we see the access list filtered out the pings to the home network.

```
R1#show access-lists LAB_BLOCK_HOME
Extended IP access list LAB_BLOCK_HOME
    10 deny ip 10.10.10.0 0.0.0.255 10.0.0.0 0.0.0.255 log (8 matches)
    20 deny ip 10.10.20.0 0.0.0.255 10.0.0.0 0.0.0.255 log
    30 deny ip 10.10.99.0 0.0.0.255 10.0.0.0 0.0.0.255 log
    40 permit ip any any
R1#show logging | i IPACCESSLOG
*Mar  1 13:49:10.264: %SEC-6-IPACCESSLOGDP: list LAB_BLOCK_HOME denied icmp 10.10.10.21 -> 10.0.0.1 (0
/0), 1 packet
R1#
```

Test 9: Second device on a secured port (optional)

What it proves: Port security
Pass criteria: The violation is detected and logged
Result: Pass — %PORT_SECURITY-2-PSECURE_VIOLATION naming the second MAC; port err-disabled and recovered with shutdown / no shutdown

Procedure and evidence

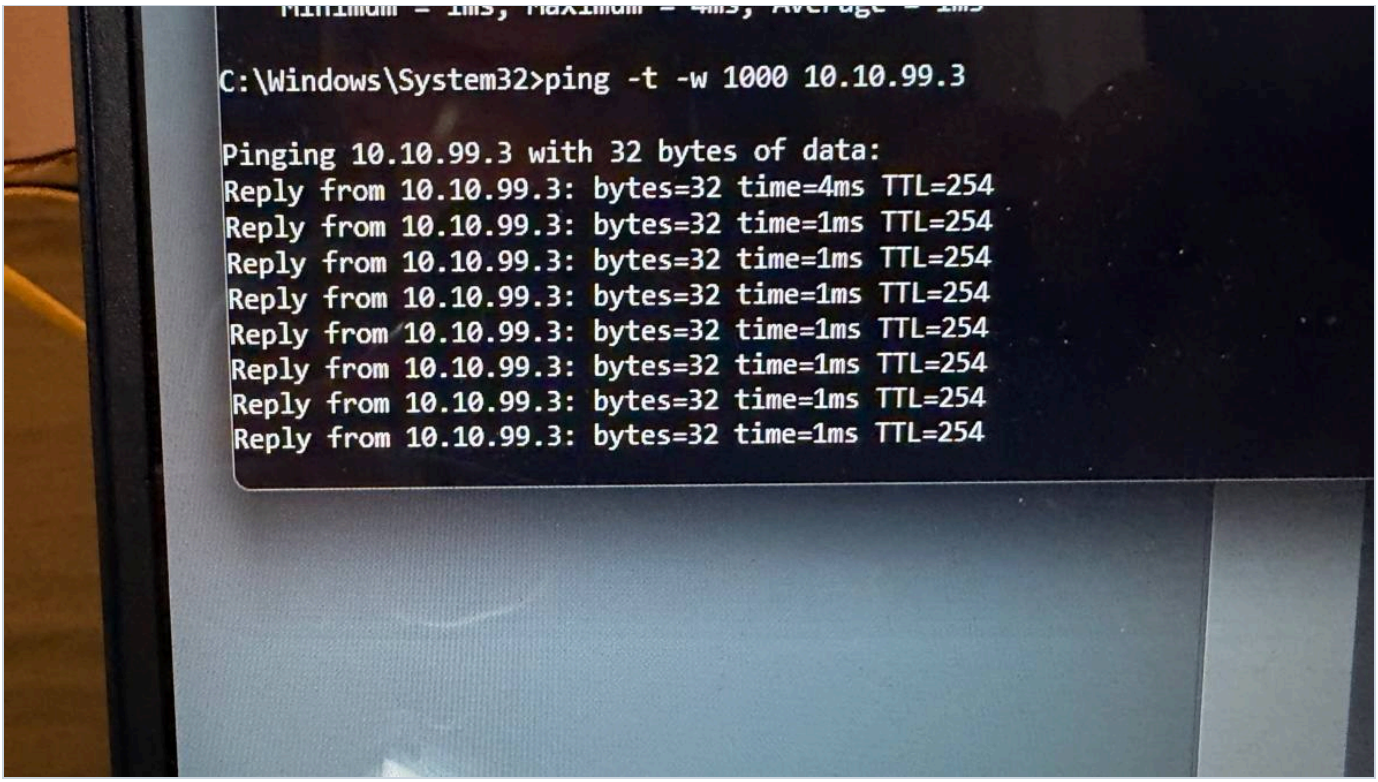
Initially lets see the port security on the interface the test laptop is connected to on SW2's g0/3:

```
SW2#sh port-security int g0/3
Port Security                  : Enabled
Port Status                    : Secure-up
Violation Mode                 : Shutdown
Aging Time                     : 0 mins
Aging Type                     : Absolute
SecureStatic Address Aging     : Disabled
Maximum MAC Addresses          : 1
Total MAC Addresses            : 1
Configured MAC Addresses       : 0
Sticky MAC Addresses           : 1
Last Source Address:Vlan       : 0800.2749.db1d:10
Security Violation Count       : 0

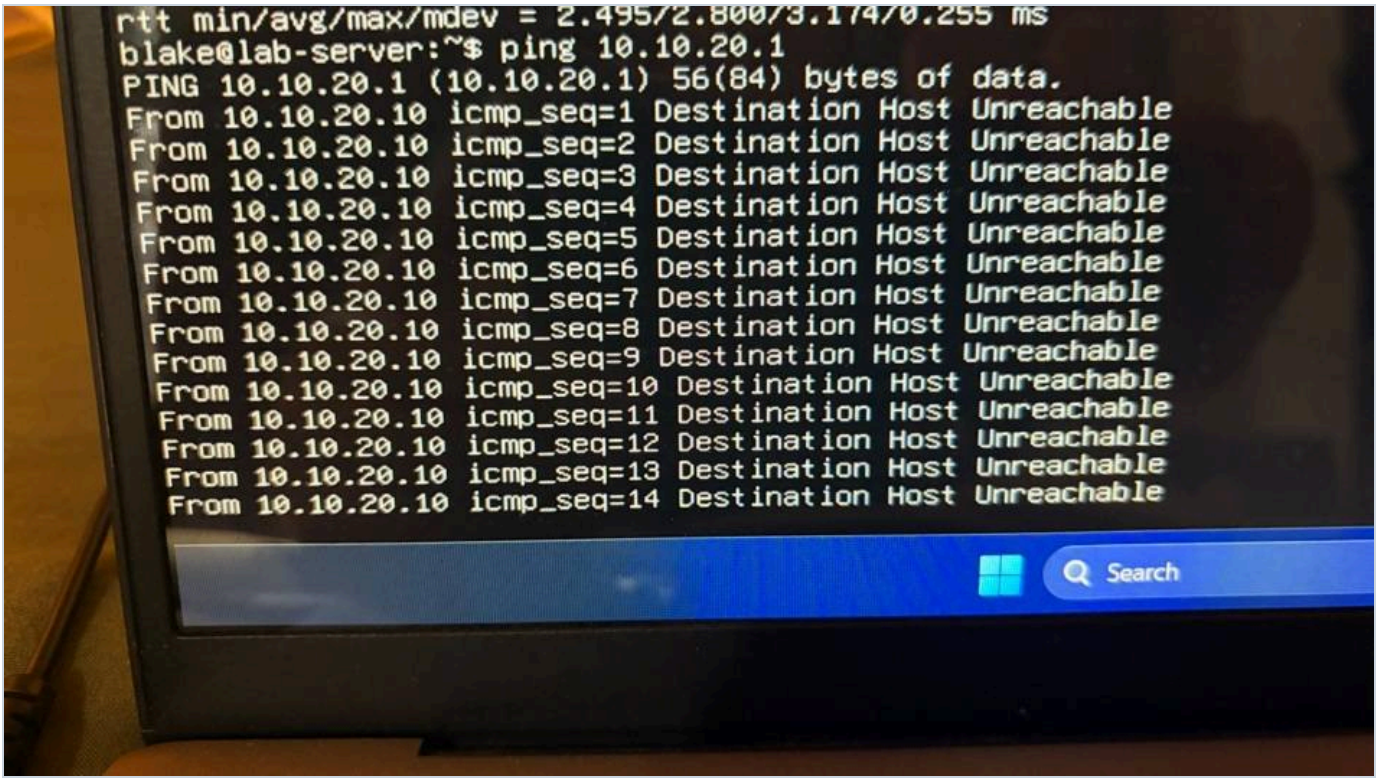
SW2#
```

Now to test I will ping on the test laptop:
"ping -t -w 1000 10.10.99.3" then ping on the ubuntu server I will ping "ping 10.10.99.3"

On Windows laptop:



On SRV1:



We see SRV1 can’t ping an IP it has been before since the port it’s own only allows 1 mac address. It learned it from my windows computer first and is not allowed to learn a second device, being the ubuntu server.
We see the port is not in secure-shutdown mode and see it got an increased violation.

```
SW2#show mac address-table interface g0/3
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
SW2#sh port
SW2#sh port-security int g0/3
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0800.2749.db1d:10
Security Violation Count   : 1

SW2#show logging | i PSECURE|ERR_DISABLE
*Mar  1 05:26:30.721: %PM-4-ERR_DISABLE: psecure-violation error detected on Gi0/3, putting Gi0/3 in e
rr-disable state
*Mar  1 05:26:30.729: %PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC a
ddress 0800.2749.db1d on port GigabitEthernet0/3.
SW2#
```

To recover the port, I turned off the VM and when into g0/3 and re enabled the port via:
int g0/3
shut
no shut

```
SW2(config)#int g0/3
SW2(config-if)#shut
SW2(config-if)#no shut
SW2(config-if)#
SW2(config-if)#end
SW2#sh por
*Mar  1 05:34:40.112: %SYS-5-CONFIG_I: Configured from console by blake on console
*Mar  1 05:34:40.548: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed stat
e to up
SW2#sh port
SW2#sh port-security
*Mar  1 05:34:41.873: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
SW2#sh port-security int g0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0800.2749.db1d:10
Security Violation Count   : 0

SW2#
```

Now you can see g0/3 is back up.